CCF501 Cloud Computing Fundamentals

CCF501_Madipoju_Prathyusha_M_Technology Report and Presentation

Assessment – 1

Technology Report and Presentation

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Application

To guarantee scalability and manage greater demand during peak hours, ABC Enterprise should host its web applications and servers on (EC2). This will ensure that your data is available and secure.

Models for Cloud Services

There are three primary service models for cloud computing services: Infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

IaaS: IaaS offers storage, networks, virtual machines, and other virtualized computer resources over the Internet (Hanczewski & Stasiak, 2024). These virtualized resources may execute the operating systems and programs the users have installed.

PaaS: With PaaS, users can create, deploy, and manage apps without worrying about the nitty-gritty of the underlying Infrastructure (Isharufe & Jaafar, 2020). This category includes databases, middleware, and integrated development environments (IDEs).

SaaS: Subscription-based SaaS provides access to various applications over the Internet. The cloud provider handles everything, from the underlying Infrastructure to the application software and data.

Cloud Deployment Frameworks

The models used to deliver cloud services outline the steps consumers must take to access these services. Here are the four most common deployment models:

The public cloud: Multiple companies share the same public cloud services, which are offered via the public Internet (Deng & Qin, 2020). In the public cloud, may find services provided by companies like AWS, Google Cloud, and Microsoft Azure.

A Private Cloud: Dedicated to a single enterprise, private cloud services run on a private network that may be hosted in-house or by an outside supplier (Dhaya & Kanthavel, 2021).

Hybrid Cloud: With a hybrid cloud, users are able to access and share resources across both public and private clouds (Lei & Wu, 2022).

Multi-Cloud: To avoid becoming too reliant on any one vendor and make the most of the capabilities offered by each platform, multi-cloud uses several distinct cloud services from several suppliers.

Proper ABC Enterprise Models

The best models to meet ABC Enterprise's needs are: IaaS is the service model.

Rationale: By using Infrastructure as a Service, ABC can manage and configure its servers, storage, and networks with the required control and flexibility.

Deployment Model: Public Cloud

Rationale: An important consideration for ABC, a startup, is the low cost of the public cloud architecture. Public clouds, like as AWS, provide excellent scalability and dependability, guaranteeing that ABC can manage demand fluctuations without experiencing downtime.

Problem Statement

Several serious IT issues ABC Company's expansion and productivity. Startups in particular may find it difficult to afford the large initial expenditures required by traditional IT infrastructure, which include hardware, software, and physical space. There are performance issues during peak hours since this design isn't scalable enough to handle the growing number of customers. The lack of thorough disaster recovery plans and redundant systems makes it even more difficult to maintain high availability and dependability, leading to frequent interruptions and information security issues. Due to its rigidity, the traditional paradigm needs continual human involvement to adjust resource allocations in response to unpredictable workload fluctuations. All of these problems make it harder for ABC Enterprise to be innovative, quick to react to customer needs, and reliable in their service. A more adaptable, scalable, and economical IT infrastructure solutions is a need for ABC Enterprise if it wants to keep up with the competition and make the most of its resources.

Steps to Overcome

To solve their IT problems, ABC Enterprise moved to cloud computing, which drastically cut their initial investment. The company could manage higher demand during peak hours by using (EC2) for versatile and scalable hosting of web app servers.

Route53

Pricing for Route 53 on Amazon A DNS web service that is both scalable and very available is Route 53. Some important parts of the cost are:

- Prices for hosted zones start at $0.50/month for the first 25 zones and drop to $0.10/zone after that (Haque, 2020).
- DNS costs $0.40 per million for the first one billion enquiries, and for every billion queries after that, it drops to $0.20.
- Monthly health checks cost $0.50 each.

Costs of Amazon S3

Scalable object storage is available with Amazon S3. Aspects of the cost consist of:

- Standard Storage costs $0.023 per GB up to 50 TB per month, while Infrequent Access costs $0.0125 per GB
- The cost for each 1,000 requests is $0.005 for PUT, COPY, POST, and LIST operations, and $0.0004 for GET operations.
- For data transfers up to 10 TB per month, the cost is $0.09 per GB.

Elastic Compute Cloud at Amazon

There are two main pricing structures for the resizable computational capability that EC2 offers:

- Ideal for unexpected workloads, on-demand instances allow to pay by hour or second with no long-term obligations (Choudhary A. , 2021).
- Save as much as 75% with a one- or three-year commitment; this plan works well for workloads that remain relatively constant

Suitable Cost Model for ABC Enterprise

- Route 53: An Appropriate Cost Model for ABC Enterprise Make sure dependability and performance are guaranteed by using standard pricing.
- Amazon Simple Storage Service (S3): For data that is accessed often, choose Standard Storage. For data that has to be accessed infrequently, use Infrequent Access.
- Amazon Elastic Compute Cloud: To maximize efficiency and minimize expenses, use a mix of On-Demand Instances for unpredictable, short-term demands and Reserved Instances for more predictable, long-term workloads.

Best Cloud AWS

Because of its maturity and extensive service offerings, AWS is perfect for innovation and adaptability. Azure is ideal for businesses that use Microsoft products since it interfaces with them smoothly. Applications that rely heavily on data will find GCP's strengths in machine learning and analysis to be an ideal fit.

Elastic Compute Cloud (EC2) by Amazon

Description: Amazon EC2 offers the elastic cloud computing capacity. Users may start instances, which are virtual servers, with different combinations of compute, memory, storage, and networking capabilities.

Key Features:

- A vast array of instance types are designed for certain use cases, including instances optimized for computation, memory, and storage, among others (Choudhary & Verma, 2021).

- Elasticity is the capacity to autonomously scale up or down using Auto Scaling groups, guaranteeing the best possible performance and cost-effectiveness.
- On-demand, Reserved, and Spot Instances are some pricing choices that may be used to minimize costs according to consumption patterns.

Application for ABC Company:

ABC may host their app's web and application servers on Amazon EC2 instances. The scalability feature guarantees the service can handle increased demand without interruption at peak periods, like a successful marketing campaign.

Amazon Simple Storage Service (S3)

Description: Amazon S3 is an object storage solution that can store and retrieve unlimited data from any location on the web. It is also very secure and has a long lifespan (Teylo & Brum, 2020). Regarding AWS services, S3 is well-known for its stability, high availability, and integration options.

Key Features:

- Data is securely stored and easily available with S3's availability over a year and durability.
- Automatically handles increasing data loads without sacrificing performance; scalable.
- Data management lifecycle rules may be defined, allowing for actions like automatically removing objects after a certain time or moving data to alternative storage types.

Application for ABC Company:

A lot of the static stuff ABC's software needs, such as backups and images, may be stored in S3. Durability and availability of the service guarantee that vital data is accessible and secure at all times, while lifecycle rules aid in cost management for storage by moving older data to less expensive storage classes.

Implementation

By moving their operations to the cloud, ABC Enterprise may take advantage of lower initial investment, more scalable resources to meet future expansion, and more dependability with less downtime.

Efficient use of resources

Advantages: The initial expenses of information technology infrastructure may be reduced, which is one of the benefits of cloud computing. It is possible that traditional information

technology setups need significant expenditures in hardware, software, and facilities, which may be prohibitive for startups. There is a usage-based pricing model for cloud services, which are pay-as-you-go.

Clouds have the following characteristics:

- The pay-as-you-go model through pricing allows ABC to pay only for the resources it utilizes, eliminating significant up-front costs and boosting the efficiency of its funds (Daum & Zhang, 2023).
- Elasticity guarantees that ABC will only pay for what it needs since resources may be dynamically scaled up or down in response to demand, maximizing cost efficiency.

Scalability and adaptability

Advantages: The scalability offered by cloud services is unparalleled, making them an indispensable instrument for growing companies. Cloud services are easily scalable, which allows them to accommodate ABC's expanding customer base.

Clouds have the following characteristics:

- Assuring optimal performance and cost-efficiency, elastic computing resources on cloud systems such as XYZ Cloud may scale up to meet peak demand and down when demand declines. This allows maximum resources to be used (Li & Lu, 2021).
- The growth of ABC is made possible by auto-scaling, which provides management of fluctuating workloads without requiring operator involvement.

Reliability and abundance of availability

Advantages: XYZ Cloud offers cloud services that are trustworthy and accessible, with the goal of both minimizing downtime and ensuring that applications are always available.

Clouds have the following characteristics:

- Cloud service providers provide high availability by using redundant systems and replicating data across several locations with multiple copies (Souza & da Costa, 2020).
- Cloud providers provide database administration, backup solutions, and disaster recovery plans, all of which contribute to increased dependability and reduce the work the IT management team at ABC must do.

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